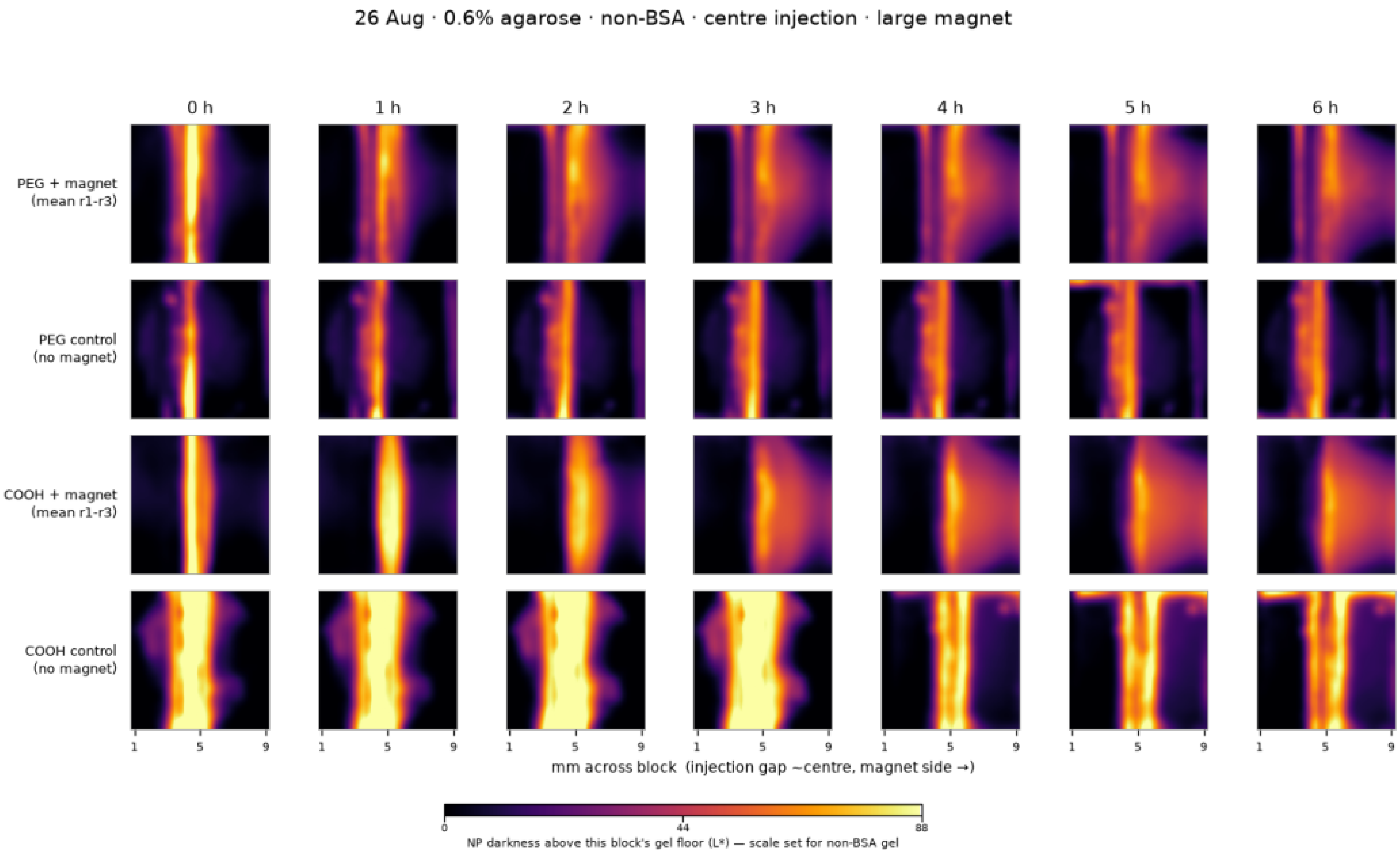
Centre-Injection Nanoparticle Transport

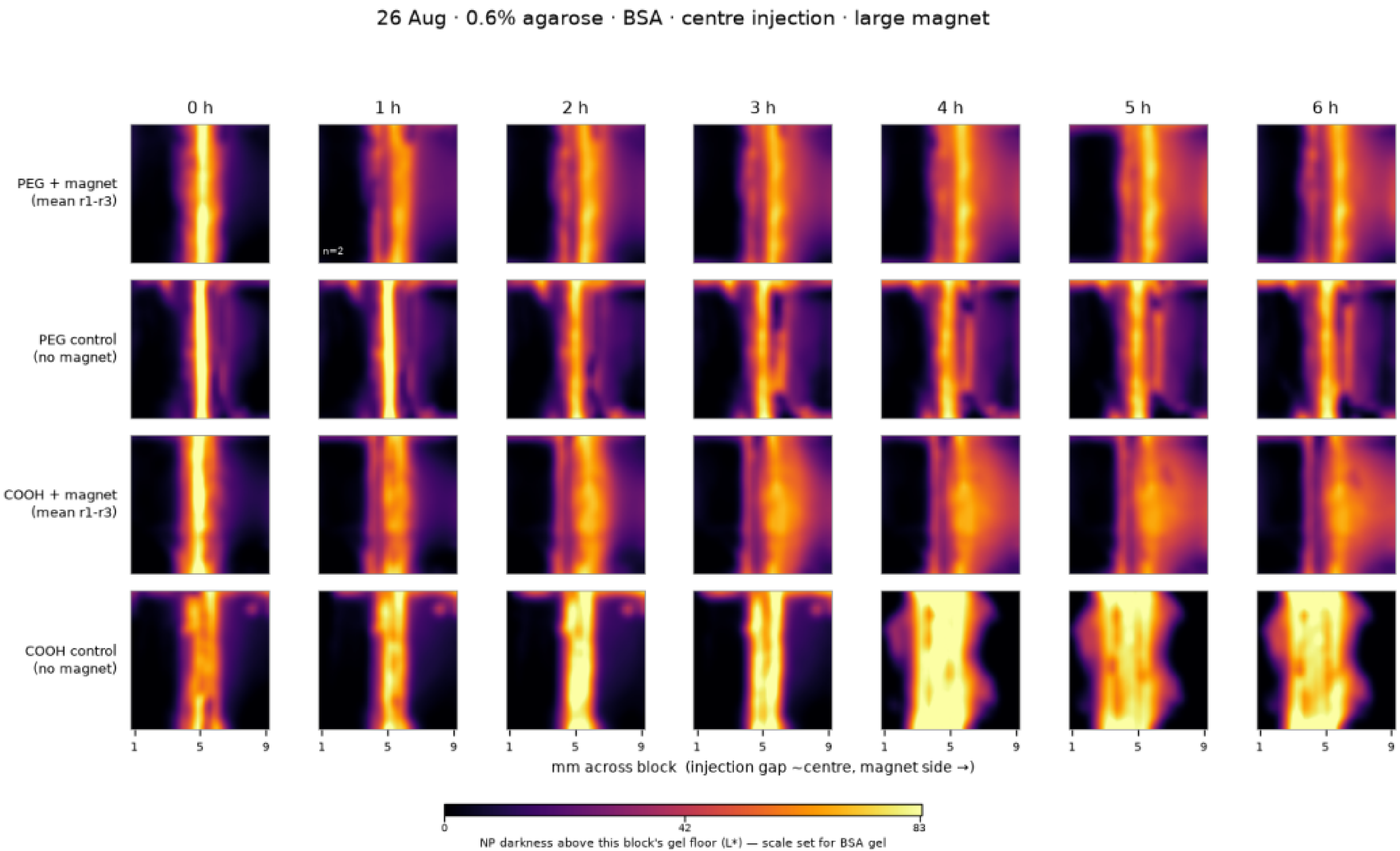
Block heatmaps — 6 h run, 26 Aug 2026 (0.6% agarose, large magnet)

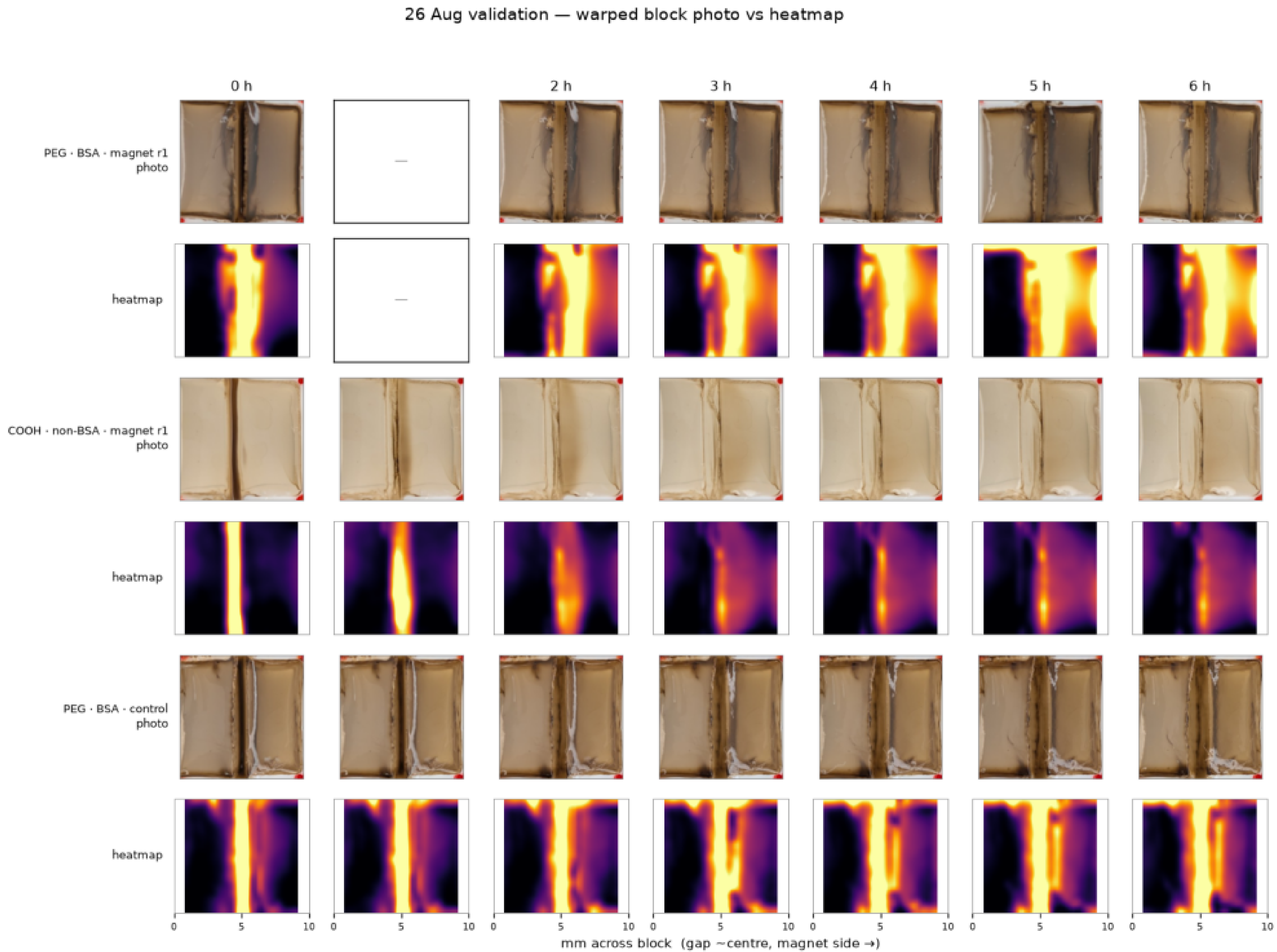
One page per agarose type (non-BSA, BSA). Rows: PEG + magnet (mean of r1-r3), PEG control, COOH + magnet (mean of r1-r3), COOH control. Columns step through the run. Each cell is the 10x10 mm block from above, spanning the whole block from its left edge with the magnet side at the right, so the centre injection gap sits near 5 mm. Block corners are read from the four red dots on the holder (208/208 frames, aspect 1.06).

Scales differ between the two panels: BSA agarose is intrinsically cloudier and darker (raw ~127 vs ~67 L*), so absolute brightness is not comparable across panels — within a panel it is. In BSA gel the cloudiness is strong enough that the controls look almost as bright as the magnet arms; the magnet-side-minus-far-side measure below cancels that and is the trustworthy readout. All four magnet arms rise monotonically (to 14-37 L* by 6 h); no control shows a monotonic trend.

github.com/sthoa/heatmaps


26 Aug · 0.6% agarose · non-BSA · centre injection · large magnet
0 h
1 h
2 h
3 h
4 h
5 h
6 h
PEG + magnet
(mean r1-r3)
PEG control
(no magnet)
COOH + magnet
(mean r1-r3)
COOH control
(no magnet)
1
5
9
mm across block (injection gap ~centre, magnet side →)
0
44
88
NP darkness above this block's gel floor (L*) — scale set for non-BSA gel


26 Aug · 0.6% agarose · BSA · centre injection · large magnet
0 h
1 h
2 h
3 h
4 h
5 h
6 h
PEG + magnet (mean r1-r3)
PEG control (no magnet)
COOH + magnet (mean r1-r3)
COOH control (no magnet)
n=2
1
5
9
mm across block (injection gap ~centre, magnet side →)
0
42
83
NP darkness above this block's gel floor (L*) — scale set for BSA gel


26 Aug validation — warped block photo vs heatmap
0 h
2 h
3 h
4 h
5 h
6 h
PEG · BSA · magnet r1 photo
heatmap
COOH · non-BSA · magnet r1 photo
heatmap
PEG · BSA · control photo
heatmap
0
5
10
mm across block (gap ~centre, magnet side →)